

System Test Plan

Instructions. In this section, you must provide your system test plan with at least 5 test cases. **If you want to provide more than 5 test cases, add an appendix at the end of this document with the 6th, 7th, etc. test cases so that page numbers for all sections match the required template! Only the first 5 test cases will be graded.**

Make sure:

- You provide your sample test data
- Test IDs are uniquely identified and descriptive
- Test descriptions are fully specified with complete inputs, specific values, and preconditions
 - Be sure to provide SPECIFIC INPUTs and VALUEs so that your test cases are repeatable
- Expected results are fully specified with specific output values
- All tests cover scenarios based on the problem statement
- All tests cover unique scenarios for the system
- All strategies for system testing are demonstrated in the tests (testing equivalence classes, testing boundary values, testing exceptions/unexpected inputs)

Test Data:

For my system test cases, we will use the following test files named `landmark.txt` and `trail.txt`

`landmark.txt:`

`LANDMARK_ID,DESCRIPTION,TYPE`

`L01,Park Entrance,Location`

`L02,Entrance Fountain,Fountain`

`L03,Waste Station 1,Pet Waste Station`

`L04,Entrance Restrooms,Restroom`

L05,Overlook 1,Overlook

L06,Rock Formation 1,Rock Formation

L07,Overlook 2,Overlook

L08,Overlook Restrooms,Restroom

L09,Waste Station 2,Pet Waste Station

L10,Hidden Gardens,Gardens

L11,Campsite 1,Campsite

L12,Campsite Restrooms,Restroom

trail.txt:

LANDMARK_ID, LANDMARK_ID, DISTANCE

L01,L02,3013

L01,L03,1046

L01,L04,1179

L02,L10,3613

L03,L05,4204

L04,L09,2311

L05,L06,1039

L06,L07,2912

L07,L08,1891

L11,L12,1066

Test ID	Description	Expected Results	Actual Results
Test #1 testStartTrailManager [UC1] (Equivalence class - loading landmarks and trails from the files)	Preconditions: - TrailUI has been loaded successfully - User starts the software and knows the location of the input files - The trail.txt and landmark.txt files exist and are loaded Steps: - Click "Get landmark data" - Browse to select the files landmark.txt - Click Done - Click on "Get trail data" - Select the trail.txt file - Click Done	The trail guide map loads and the UI displays the information in the input files: LANDMARK_ID, DESCRIPTION, TYPE L01, Park Entrance, Location L02, Entrance Fountain, Fountain L03, Waste Station 1, Pet Waste Station L04, Entrance Restrooms, Restroom L05, Overlook 1, Overlook L06, Rock Formation 1, Rock Formation L07, Overlook 2, Overlook L08, Overlook Restrooms, Restroom L09, Waste Station 2, Pet Waste Station L10, Hidden Gardens, Gardens L11, Campsite 1, Campsite	--- Displaying File: input/landmark.txt --- LANDMARK_ID,DESCRIPTION,TYPE L01,Park Entrance,Location L02,Entrance Fountain,Fountain L03,Waste Station 1,Pet Waste Station L04,Entrance Restrooms,Restroom L05,Overlook 1,Overlook L06,Rock Formation 1,Rock Formation L07,Overlook 2,Overlook L08,Overlook Restrooms,Restroom L09,Waste Station 2,Pet Waste Station L10,Hidden Gardens,Gardens L11,Campsite 1,Campsite L12,Campsite Restrooms,Restroom --- End of File ---

		L12, Campsite Restrooms, Restroom	--- Displaying File: input/trail.txt --- LANDMARK_ID,LAND MARK_ID,DISTANCE L01,L02,3013 L01,L03,1046 L01,L04,1179 L02,L10,3613 L03,L05,4204 L04,L09,2311 L05,L06,1039 L06,L07,2912 L07,L08,1891 L11,L12,1066 --- End of File ---
Test #2 testFirstAidStations [UC2] (Exception and Boundary Value - loading the paths possible with first aid stations)	Preconditions: - TrailUI has been loaded successfully - The trail.txt and landmark.txt files exists and is loaded Steps: - Click “view report of first aid stations” - Enter 0 for “the minimum amount of intersecting paths” - Click done	A popup appears with the following message: “Invalid number entered. Value must be greater than 0.”	The console prints the following: “Number of intersecting trails must be greater than 0.”

Test #3 testDistanceToReachableLandmarks[UC3] (Equivalence class - viewing	Preconditions: - TrailUI has been loaded successfully - The trail.txt and landmark.txt files exists Steps:	The report shows: Landmarks Reachable from Entrance Fountain (L02) Entrance Fountain (L02) : 0 feet Park Entrance (L01) : 3013 feet Hidden Gardens (L10) : 3613 feet	Landmarks Reachable from Entrance Fountain (L02) { 3013 feet to Park Entrance (L01) 4059 feet to Waste Station 1 (L03) 4192 feet to Entrance Restrooms (L04)

the distances from the landmark chosen)	1. Click on “produce report of distances to landmarks” 2. Enter a L02 as the ID of a landmark to be the starting point 3. Click done	Waste Station 1 (L03) : 4059 feet Entrance Restrooms (L04) : 4192 feet Waste Station 2 (L09) : 6503 feet (1.23 miles) Overlook 1 (L05) : 8263 feet (1.56 mile) Rock Formation 1 (L06) : 9302 feet (1.76 miles) Overlook 2 (L07) : 12214 feet (2.31 miles) Overlook Restrooms (L08) : 14105 feet (2.67 miles)	8263 feet (1.56 miles) to Overlook 1 (L05) 9302 feet (1.76 miles) to Rock Formation 1 (L06) 12214 feet (2.31 miles) to Overlook 2 (L07) 14105 feet (2.67 miles) to Overlook Restrooms (L08) 6503 feet (1.23 miles) to Waste Station 2 (L09) 3613 feet to Hidden Gardens (L10) }
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Test #4 testCloseTrail Manager [UC4] (Equivalence Class - closing the trail manager)	Preconditions: • Test #1 has passed Steps: 1. Click on “quit” to close the TrailManager	The UI closes and the input files remained unchanged.	UI closes and console prints Exiting TrailManager. Goodbye!
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Test #5 testNoLandmarkToFindDistance (Exception/unexpected input - there is not a landmark with the given ID)	Preconditions: • TrailUI has been loaded successfully • The trail.txt and landmark.txt files exists Steps: 1. Click on “produce report of distances to landmarks” 2. Enter a ID of a landmark to be the starting point that does not exist 3. Click done	A error message pops up which says “There is no Landmark with the following ID”	The following message popups: The provided landmark ID (L30) is invalid for the park.
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